

AE4SR - Seminary Work Assignment

Part I. – Input-output analysis and synthesis of a linear time-invariant continuous-time system

Consider a system described by the following ordinary differential equation:

$$a_2 y''(t) + a_1 y'(t) + a_0 y(t) = b_1 u'(t) + b_0 u(t)$$

with particular parameters.

Solve the following tasks (numbers in square brackets mean the maximum possible score):

1. Find the corresponding transfer function. [0.25]
2. Calculate poles, zeros and the static gain, and decide about the stability and periodicity. [0.75]
3. Calculate the impulse response and draw its plot by using a suitable SW. [1.25]
4. Calculate the step response and draw its plot by using a suitable SW. [1.25]
5. Calculate the corresponding frequency transfer function expressed in both the algebraic (real&imaginary component) form and goniometric (exponential) form. [1]
6. Choose at least 10 frequency values and insert them into the algebraic form of the frequency transfer function (i.e., calculate the function values). Display the results in a table and also as the Nyquist plot. [1.25]
7. Choose at least 10 frequency values and insert them into the goniometric form of the frequency transfer function (i.e., calculate the function values). Display the results in a table and also as Bode plots. [1.25]
8. Let the simple negative feedback control system (1DoF) be with a step-wise reference and a step-wise load disturbance signals. By using the polynomial algebraic method, design a controller satisfying control feedback stability, (asymptotic) reference tracking, and disturbance rejection. Let feedback poles be set as $-a_0, -a_1, -0.5(a_0 + a_1)$. Display control responses $u(t), e(t), y(t)$ (by using Matlab). [2]
9. Design the PI(D) controller via another (arbitrary) simple control design method (e.g., using the model step response characteristics). Display control responses $u(t), e(t), y(t)$ (by using Matlab). [1.5]
10. Verify control feedback asymptotic (Lyapunov) stability for control systems from tasks #8 and #9 by using an arbitrary stability criterion. [0.5]

Part II. – State-space model description, solution, and synthesis for a linear time-invariant continuous-time system

11. Find two different state-space descriptions of the system model. [1.5]
12. For both the state-space descriptions, perform (reversely) the input-output differential equation (or transfer function) calculation. [1]
13. Verify that both the state-space descriptions are controllable and observable. [0.5]
14. Calculate the standard fundamental matrix for one (selected) state-space description from task #11. [1.5]
15. Solve the state-space equation for zero initial conditions and $u(t) = 1(t)$ (by means of the standard fundamental matrix). Then calculate the output and compare the result with that from task #4. [2]
16. For a suitable state-space model from task #11, design a proportional state-feedback controller satisfying control feedback stability and (asymptotic) reference tracking. Let feedback poles be set as $-a_0, -a_1, -0.5(a_0 + a_1)$. Display control responses $u(t), e(t), y(t)$ (by using Matlab). [2]
17. Compare (in words) the obtained results from tasks #8, #9, and #16. [0.5]

Total points: 20

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